

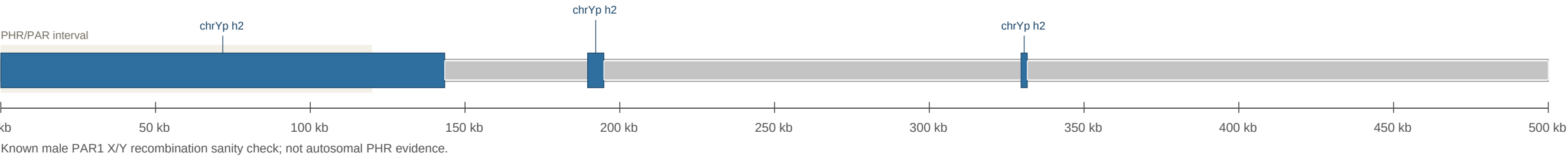
Candidate Fig. 5 asset pack: PAR1 control and autosomal PHR-compatible pedigree patches

Existing WashU odgi-untangle patch calls; same-chromosome background is subdued and non-homologous donor arms are semantically highlighted.

A PAR1 positive control: paternal chrX p maps to chrY p

PAN027 paternal (hap2) vs PAN011 (father)

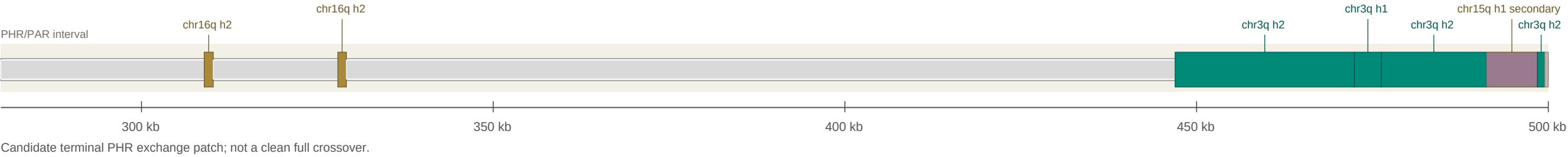
Main chrYp block: 0-143,431 bp; mean/min score 0.9998/0.9933; C15/C15.



B PAN027 autosomal candidate: chr9 q terminal patch mostly chr3 q

PAN027 paternal (hap2) vs PAN011 (father)

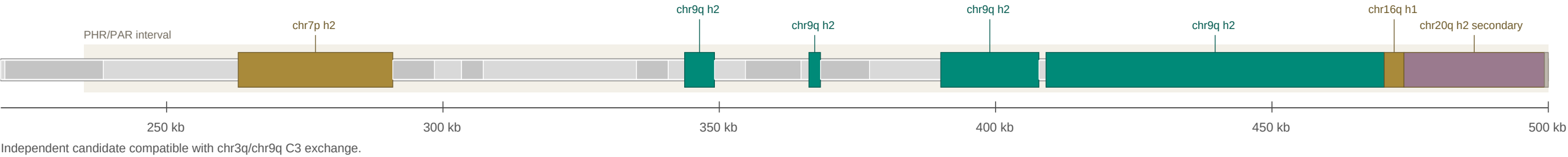
Terminal switch after 446.9 kb is mostly chr3q; chr15q is cross-community secondary; chr20q tail is low-confidence.



C Independent PAN028 support: chr3 q terminal patch maps to chr9 q

PAN028 maternal (hap1) vs PAN027 (mother)

Strong chr9q blocks: 390,073-407,837 and 409,115-470,323 bp; C3/C3.



D Selected finite candidate set

Deliberately selected review set; not a genome-wide rediscovery run.

panel	role	primary evidence	secondary	low-confidence
A	PAR1 positive control	chrYp 0-143431 (min 0.9933, within_community); chrYp 189599-194850 (min 0.9998, within_community); chrYp 329647-331663 (min 1.0000, within_community)	none emphasized	none emphasized
B	PAN027 chr9q candidate	chr3q 446944-472441 (min 0.9963, within_community); chr3q 472441-476266 (min 0.9320, within_community); chr3q 476266-491186 (min 0.9768, within_community); chr3q 498459-499507 (min 0.9801, within_community)	chr16q 308916-310201 (min 1.0000, within_community); chr16q 327917-329124 (min 1.0000, within_community); chr15q 491186-498459 (min 0.9975, cross_community)	chr20q 499507-500000 (min 0.8577, cross_community)
C	PAN028 chr3q candidate	chr9q 343727-349132 (min 0.9994, within_community); chr9q 366221-368310 (min 0.9701, within_community); chr9q 390073-407837 (min 0.9991, within_community); chr9q 409115-470323 (min 0.9909, within_community)	chr7p 262953-290922 (min 0.9432, within_community); chr16q 470323-473889 (min 0.9969, within_community); chr20q 473889-499310 (min 0.9820, cross_community)	chr9q 499310-500000 (min 0.5545, within_community)

same-chromosome background PAR1 chrY donor chr3/chr9 candidate donor secondary/cross-community low-confidence tail